

RUVA — Project Requirements Document (PRD)

Version 1.2 · Flutter Mobile Application · Enhancement of Existing Application

1. Project overview

RUVA is a centralised personal health, fitness and wellness application created by Chef Ruu (Ruvarashe).

The application will expand the existing Fasting Companion app into an all-in-one system where users can track fasting sessions, record food and nutrition, manage kitchen ingredients, discover recipes, log exercise and hydration, and monitor selected health trends.

The application will be built using Flutter and Dart, extending the existing codebase rather than rebuilding it from scratch.

Product vision

To provide one convenient place to understand and manage everyday wellness habits, helping users make informed decisions about food, movement, fasting and personal health.

Core objectives

- Preserve the existing circular fasting countdown and milestone interface.
- Introduce a food diary and nutrition tracking.
- Allow users to maintain a reusable inventory of ingredients in their kitchen.
- Recommend recipes based on available ingredients and nutritional information.
- Track exercise, water intake and other selected wellness habits.
- Provide personal history, charts and summaries.
- Support monthly kitchen inventory updates.
- Keep the interface clean, accessible and easy to use.

2. Existing application functionality

RUVA v1.2 must build upon the current application and preserve its working functionality.

2.1 Circular fasting timer

The main Timer screen must retain the existing circular countdown design.

It should display:

- Elapsed fasting time.
- Planned fasting duration.
- Remaining time.
- Session start time and planned end time.
- Clear Start Fast and End Fast controls.
- Active session status.
- Access to previous fasting sessions.

The timer must continue working when the application is closed and reopened. Elapsed time must be calculated from saved timestamps rather than relying solely on a running counter.

2.2 Fasting milestone indicators

The circular ring will display milestone icons representing educational information about changes that may occur during fasting.

- Yellow: estimated fat-burning milestone.
- Red: estimated ketosis milestone.
- Light blue: primary interface and timer design.

Users should be able to tap a milestone to read an explanation.

Important: milestone timing varies between individuals. The application must not claim that elapsed time confirms ketosis, fat burning, autophagy or any other specific internal metabolic state.

2.3 Interface and navigation

The interface must retain its light-blue visual identity while reducing clutter.

Proposed main navigation:

Section	Purpose
Timer	Fasting countdown and milestones
History	Past fasting sessions and recorded activities
Nutrition	Food diary, kitchen inventory, recipes and nutrition
Settings	Profile, preferences, reminders and privacy

The fasting timer must remain the application's primary screen. Additional functionality should be organised into secondary sections rather than crowding the countdown.

3. New feature: Food diary and nutrition tracking

RUVA must allow users to record their meals and understand the nutritional information available for the foods they consume.

Functional requirements

1. Users can add breakfast, lunch, dinner and snacks.
2. Users can record food quantities, serving sizes and meal times.
3. Users can search a food database.
4. Users can save frequently consumed foods and meals.
5. Users can create their own food entries when an item is unavailable.
6. Users can view estimated nutrition information.
7. Users can review daily food records and previous meals.
8. Users can optionally log a meal directly from a recipe.

Nutrition information should include, where available:

- Energy.
- Protein.
- Carbohydrates.
- Fats.
- Fibre.
- Selected vitamins and minerals.

Food information must clearly indicate when values are estimated or unavailable. Database selection must consider licensing, attribution, geographic coverage and cost.

Optional future features

- Barcode scanning.
- Photo-assisted food recognition.
- Voice-based food logging.
- AI-assisted meal entry.

AI-generated entries must be reviewed by the user before being saved.

4. New feature: My Kitchen — ingredient inventory

This is one of RUVA's central new features.

Users must be able to record the raw ingredients they have at home and update their inventory as ingredients are purchased, used or discarded.

4.1 Ingredient categories

Category	Examples
Protein-rich foods	Eggs, chicken, fish, beans, lentils, tofu
Carbohydrate-rich foods	Rice, oats, potatoes, pasta, maize meal
Vegetables	Spinach, cabbage, carrots, tomatoes
Fruits	Apples, bananas, oranges
Fats, nuts and seeds	Avocado, nuts, seeds, cooking oils
Dairy and alternatives	Milk, yoghurt, cheese, fortified alternatives
Herbs and spices	Garlic, pepper, curry powder, herbs
Pantry essentials	Flour, canned tomatoes, stock, canned beans

Categories are for convenient browsing. Each ingredient may contain multiple nutrients, so nutritional values must be stored separately where data is available.

4.2 Ingredient information

Each ingredient record should support:

- Ingredient name.
- Category or categories.
- Quantity available.
- Measurement unit.
- Food state, such as fresh, frozen, canned or dried.
- Optional purchase date.
- Optional expiry date.
- Optional brand and notes.
- Nutritional information or a reference to its source.

4.3 Inventory management

Users must be able to:

- Add new ingredients.
- Edit ingredient details.
- Update quantities.
- Search and filter the inventory.
- Mark ingredients as finished.

- Remove ingredients.
- Identify ingredients approaching their recorded expiry dates.
- Identify low-stock items where quantities are available.

The application must not automatically delete inventory simply because a new month begins.

5. New feature: Smart Recipe Planner

The Smart Recipe Planner will use the user's kitchen inventory to recommend practical meals.

Instead of asking users to browse hundreds of recipes, RUVA should first identify what they can make with the ingredients they already have.

5.1 Recipe recommendation logic

The system should consider:

- Ingredients available in the kitchen.
- Quantities available, where recorded.
- Ingredients approaching expiry.
- Dietary preferences.
- Recorded allergies.
- Preparation and cooking time.
- Nutritional information.
- Meal variety.
- Missing ingredients and estimated additional cost, where available.

5.2 Recipe cards

Each suggested recipe should display:

- Recipe name and image.
- Preparation and cooking time.
- Number of servings.
- Ingredients available.
- Missing ingredients.
- Cooking instructions.
- Estimated nutrition information.
- Dietary labels where supported.

Users should be able to save favourite recipes, adjust servings, view substitutions and mark recipes as cooked.

5.3 Ingredient matching

The system should prioritise recipes that use ingredients already available.

For example, if a user has eggs, rice, spinach, tomatoes and onions, RUVA might suggest:

- Spinach and tomato scrambled eggs.
- Vegetable and egg rice.
- A vegetable-based rice bowl.

Recommendations must be based on actual inventory records and a recipe database, not invented assumptions about what the user has.

5.4 Food safety and allergies

- Never recommend ingredients that the user has explicitly excluded due to an allergy.
- Warn users that ingredient databases cannot guarantee a recipe is free from allergens or cross-contamination.
- Do not recommend using spoiled ingredients.
- Make ingredient substitutions clear.
- Label nutritional values as estimates when appropriate.

The first release should use a dependable recipe catalogue and rule-based matching. AI-generated recipes can be introduced later after ingredient validation and food-safety checks are established.

6. New feature: Monthly kitchen review

RUVA must support a recurring monthly process that keeps the kitchen inventory accurate.

Monthly workflow

1. Send an optional monthly reminder.
2. Display the user's existing ingredients.
3. Ask the user to confirm which ingredients remain.
4. Allow quantities to be updated.
5. Allow finished or spoiled ingredients to be removed.
6. Allow new purchases to be added.
7. Refresh recipe recommendations.
8. Optionally generate a shopping list for missing ingredients.

The monthly review should preserve previous meal logs and inventory history where appropriate.

Users must be able to disable reminders and choose when they receive them.

7. New feature: Goals and personal measurements

RUVA may include a section for recording selected personal measurements and wellness preferences.

Functional requirements

- Allow users to record height and weight where appropriate.
- Support periodic weigh-ins and a private history chart.
- Allow users to select general wellness goals.
- Provide editable measurement records.
- Allow users to opt out of weight tracking.
- Explain the limitations of BMI if it is displayed.
- Keep sensitive profile information optional.

BMI must not be used as the sole basis for determining health, prescribing a diet or calculating restrictive calorie targets. Adult BMI classifications should not be applied to minors. RUVA must use age-appropriate safeguards and must not generate restrictive weight-loss plans for users under 18.

The application should support general wellness and balanced nutrition rather than treating weight loss as the default objective.

8. New feature: Exercise and activity tracking

RUVA should allow users to record their physical activities and review their history.

Functional requirements

Users should be able to record:

- Activity type.
- Date and time.
- Duration.
- Optional intensity.
- Optional notes.

The application should display a history of recorded activities and simple trends over time.

Future enhancements

- Step tracking.
- Wearable integrations.
- Health-platform synchronisation.

- Smartwatch support.
- Additional activity metrics.

These integrations must be implemented only after confirming platform support, permissions, data quality and technical feasibility.

9. New feature: Hydration tracking

Users should be able to record water intake throughout the day.

Functional requirements

- Add a water entry.
- Select an amount and measurement unit.
- View a daily hydration summary.
- Review previous entries.
- Configure optional reminders.

Hydration reminders must be user-controlled. The application should not make medical claims based solely on recorded water intake.

10. History, charts and insights

RUVA should provide a central place for users to review their recorded information.

Required capabilities

- Fasting session history.
- Calendar or timeline navigation.
- Food diary history.
- Recipe history.
- Exercise history.
- Hydration history.
- Selected personal measurement history.
- Simple charts and date filters.
- Daily summaries based on recorded information.

Charts must use real recorded data, identify their date ranges and display appropriate measurement units.

The application must not invent missing data or imply that a user completed an activity that was never recorded.

11. Optional future functionality

The following features should be considered for later releases rather than included automatically in the initial v1.2 scope.

Feature	Description
AI recipe assistant	Suggest recipes based on inventory and preferences
AI wellness assistant	Answer general questions using appropriate, limited wellness guidance
Recipe import	Save recipes from supported sources
Grocery list integration	Generate shopping lists from missing ingredients
Food barcode scanning	Find packaged foods using barcodes
Photo-assisted logging	Help identify meal components from images
Wearable integrations	Import supported activity data
Health-platform synchronisation	Connect to supported platform health data
Advanced nutrition reports	Review longer-term food and nutrient patterns
Progress photos	Optional private photo records
Custom trackers	Allow selected personal metrics and habits
Professional sharing	Share selected information with a chosen professional, with consent
Family or community features	Share selected recipes or participate in moderated groups

Any external integration must be assessed for licensing, permissions, privacy, security, ongoing maintenance and potential costs before implementation.

12. Technical requirements

Frontend

- Flutter.
- Dart.

- Android and iOS support.
- Responsive layouts for supported mobile screen sizes.

Existing application

Before development begins, the existing codebase must be audited to identify:

- Current screens and navigation.
- Existing timer logic.
- State-management approach.
- Local storage and backend dependencies.
- Existing data structures.
- Dependencies and packages.
- Test coverage.
- Potential data migration risks.

The development team must preserve working functionality and avoid replacing existing architecture without a documented reason.

Architecture

The codebase should be organised into maintainable feature modules, such as:

- Timer.
- History.
- Food diary.
- Kitchen inventory.
- Recipes.
- Goals and measurements.
- Activity and hydration.
- Settings.

Existing state management and persistence should be reused if suitable.

Backend and integrations

A backend may be required for accounts, cross-device synchronisation, external food databases, AI services or third-party integrations.

API credentials and privileged secrets must not be embedded directly in the Flutter application.

Core manual logging and inventory functions should work offline where the existing architecture permits, with clear handling of unavailable online services.

13. Initial data model

The application should support the following main entities.

Entity	Purpose
UserProfile	Preferences, units, optional profile information
FastingSession	Start time, end time, planned duration and status
MilestoneConfig	Educational milestone labels, icons and display settings
Ingredient	Kitchen inventory, quantity, unit and optional expiry
FoodEntry	Food, portion, meal time and available nutrition values
Recipe	Ingredients, quantities, instructions and servings
RecipeLog	Recipes prepared and optional links to food entries
ActivityEntry	Exercise and other recorded activities
HydrationEntry	Water intake and timestamps
MeasurementEntry	Selected personal measurements and units
MonthlyInventoryReview	Review date and inventory changes

All records should have stable identifiers and appropriate timestamps. Data migrations must be tested before existing user data is changed.

14. Non-functional requirements

Performance

Screens should load promptly, and searching ingredients or filtering recipes should remain responsive as the user's data grows.

Reliability

The timer and user-entered records must persist correctly across application restarts.

Privacy

Collect only necessary personal information. Explain why sensitive data is requested and provide appropriate controls for export and deletion.

Security

Protect personal information in transit and at rest where applicable. Use secure authentication and backend controls when remote services are introduced.

Accessibility

Support screen readers, scalable text, accessible touch targets and adequate colour contrast. Information must not rely on colour alone.

Maintainability

Use reusable Flutter widgets, consistent theme tokens, testable business logic, documented dependencies and data migrations.

Compatibility

Test supported Android and iOS devices, background and resume behaviour, notification permissions, offline use and relevant screen sizes.

15. Safety and responsible health information

RUVA must follow these principles throughout the application:

1. Fasting milestones are estimates, not measurements of metabolic state.
2. Nutrition values must be identified as estimates where appropriate.
3. The application must not diagnose medical conditions.
4. BMI must not be treated as a complete measure of health.
5. The application must not encourage users to extend fasts at the expense of their wellbeing.
6. Users must be able to end fasting tracking at any time.
7. The application must not use shame, guilt or competitive pressure around food, exercise or weight.
8. Sensitive health information must be handled with appropriate privacy protections.
9. Users must be able to correct their records and control optional tracking features.

16. Acceptance criteria

RUVA v1.2 will be considered functionally ready when the following requirements have been tested:

- The existing fasting timer continues to work after the upgrade.
- The timer retains accurate elapsed time after the app is closed and reopened.

- The yellow fat-burning and red ketosis milestone icons display correctly around the timer.
- Users can add, edit, search and remove ingredients.
- Ingredient quantities and measurement units are saved correctly.
- Monthly inventory reviews update stock without deleting the entire inventory.
- Recipe recommendations reflect recorded ingredients.
- Recipes show available and missing ingredients.
- Recipe instructions and estimated nutrition are displayed where available.
- Users can optionally log a prepared recipe in their food diary without creating duplicate entries.
- Food, activity and water entries can be recorded and reviewed.
- Personal measurement tracking is optional and follows age-appropriate safeguards.
- Notifications can be disabled.
- Core screens have appropriate loading, empty, error and success states.
- Automated tests cover timer calculations, data persistence, unit conversions, recipe matching, allergy filtering and inventory updates.

17. Recommended development phases

1. Phase 0 — Audit the existing application
Review the current Flutter codebase, timer, navigation, data storage, dependencies and tests.
2. Phase 1 — Navigation and foundation
Preserve the timer, improve screen organisation and establish the new Nutrition section and data models.
3. Phase 2 — Kitchen inventory
Implement ingredient management, quantities, categories, search, expiry information and monthly reviews.
4. Phase 3 — Smart Recipe Planner
Implement recipe discovery, ingredient matching, missing ingredients, cooking instructions and serving adjustments.
5. Phase 4 — Food diary and nutrition
Implement meal logging, food search, saved foods and nutrition summaries.
6. Phase 5 — Exercise, hydration and insights
Implement activity logs, water tracking, history and basic charts.
7. Phase 6 — Personal goals and integrations
Add appropriate optional measurements and evaluate external health services.
8. Phase 7 — Testing and release
Validate data migrations, accessibility, privacy, reliability and acceptance criteria.

18. Final product definition

RUVA is a personal wellness hub built on the existing Flutter Fasting Companion application. It combines fasting-session tracking with food and nutrition logging, a reusable kitchen inventory,

ingredient-based recipe recommendations, exercise and hydration tracking, history and selected personal health insights.

The product's defining advantage is that it connects everyday habits: users can record what they have in their kitchen, discover practical meals, log what they eat, track their activity and fasting sessions, and review their recorded information in one organised application.

The interface must remain calm and uncluttered, with the fasting timer as its existing centrepiece and nutrition and wellness tools available through clearly organised sections.

Development priority: audit the existing application first, preserve all working functionality, then build the kitchen inventory and recipe recommendation workflow before expanding into advanced AI or external health integrations.